

SAT Unit 3.2

Percent

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Introduction to Percent

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Introduction to Percent

Definition: Percent means "per hundred." It represents a ratio of a number to 100.

$$\text{Percent} = \frac{\text{Part}}{\text{Whole}} \times 100$$

Example: 25 out of 100 is written as 25%.

Percent Change Formula

Formula:

$$\text{Percent Change} = \frac{\text{New Value} - \text{Original Value}}{\text{Original Value}} \times 100$$

Example: If a price increases from \$80 to \$90:

$$\frac{90 - 80}{80} \times 100 = 12.5\%$$

Key Concept: Positive percent change indicates an increase, negative indicates a decrease.

Percent Greater or Less Than

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Percent Greater or Less Than

	Translates to...	Example
$p\%$ of x	$\frac{p}{100} \times x$	15% of $x \Rightarrow \frac{15}{100} \times x = 0.15x$
$p\%$ greater than x	$\left(1 + \frac{p}{100}\right) x$	15% greater than $x \Rightarrow \left(1 + \frac{15}{100}\right) x = 1.15x$
$p\%$ less than x	$\left(1 - \frac{p}{100}\right) x$	15% less than $x \Rightarrow \left(1 - \frac{15}{100}\right) x = 0.85x$

Example: If 200 is $x\%$ greater than 50, what is x ?

$$50 \times (1 + x\%) = 200$$

Key Takeaway: If something is **50% greater than** a value, multiply by 1.5.

Example of the Percent

The number a is **70% less** than the positive number b .

The number c is **60% greater** than a .

The number c is how many times b ?

Answer

- Since a is 70% less than b , we express it as:

$$a = (1 - 0.7)b = 0.3b$$

- Since c is 60% greater than a , we express it as:

$$c = (1 + 0.6)a = 1.6a$$

- Substituting $a = 0.3b$:

$$c = 1.6(0.3b) = 0.48b$$

- Thus, the number c is **0.48** times b .

Final Answer: 0.48

Key Takeaways

- Percent represents "per hundred."
- Use the **percent change formula** to calculate increases and decreases.
- "Percent greater than" and "percent less than" refer to the original value.
- Use division to find the original value when given the increased price.

Example of the Percent

The regular price of a shirt at a store is **\$11.70**.

The sale price of the shirt is **80% less** than the regular price,
and the sale price is **30% greater** than the store's cost for the shirt.

What was the store's cost, in dollars, for the shirt?

(Disregard the \$ sign when entering your answer.)

Answer

- **Step 1: Find the sale price of the shirt**

$$\text{Sale Price} = (1 - 0.80) \times 11.70 = 0.20 \times 11.70 = 2.34$$

- **Step 2: Set up equation for the store's cost**

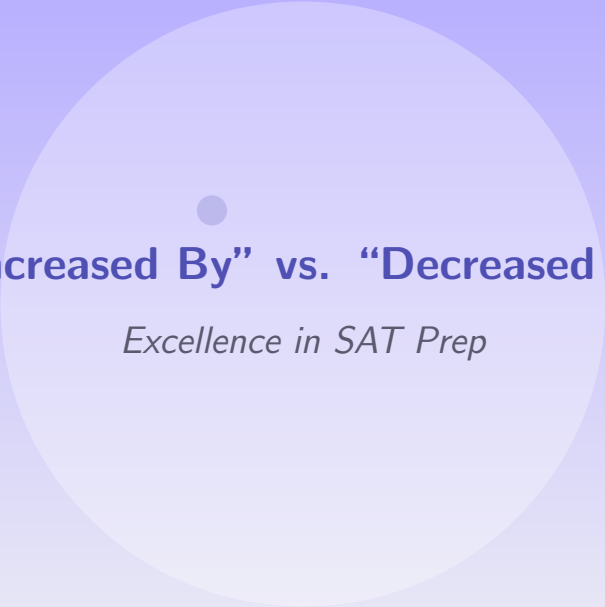
$$\text{Sale Price} = (1 + 0.30) \times \text{Cost}$$

$$2.34 = 1.30 \times \text{Cost}$$

- **Step 3: Solve for Cost**

$$\text{Cost} = \frac{2.34}{1.30} = 1.80$$

Final Answer: 1.80



“Increased By” vs. “Decreased By

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“Increased By” vs. “Decreased By”

Definitions:

- **Increased by $p\%$:** Multiply by $(1 + \frac{p}{100})$
- **Decreased by $p\%$:** Multiply by $(1 - \frac{p}{100})$

Examples:

- 120 increased by 25%:

$$120 \times \left(1 + \frac{25}{100}\right) = 120 \times 1.25 = 150$$

- 150 decreased by 20%:

$$150 \times \left(1 - \frac{20}{100}\right) = 150 \times 0.8 = 120$$

Important Notes

Important Notes:

- **Be careful:** A 20% decrease followed by a 20% increase does **not** return to the original value!

$$100 \rightarrow 80 \rightarrow 96$$

- Always treat “**increased by**” and “**decreased by**” as multiplicative — not additive.
- If you are given a final value and percentage, **reverse the multiplier** to solve for the original.

Key Insight: Think in terms of **scaling**, not just adding or subtracting!

Question: Net Percentage Increase

The value of a collectible comic book increased by 167% from the end of **2011** to the end of **2012**, and then decreased by 16% from the end of **2012** to the end of **2013**.

What was the net percentage increase in the value of the collectible comic book from the end of **2011** to the end of **2013**?

- A. 124.28%
- B. 140.28%
- C. 151.00%
- D. 209.72%

Answer and Explanation

Let the original value at the end of 2011 be 100 (for simplicity).

- After a 167% increase:

$$100 + 1.67 \times 100 = 267$$

- After a 16% decrease:

$$267 \times (1 - 0.16) = 267 \times 0.84 = 224.28$$

- Net increase from original 100:

$$224.28 - 100 = 124.28 \Rightarrow \text{Net increase is } \boxed{124.28\%}$$

Correct Answer: A. 124.28%

Practices

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Percent Greater Relationship

In 2010, the number of houses built in Town A was n percent greater than the number of houses built in Town B. If 700 houses were built in Town A in 2010, which expression represents the number of houses built in Town B in 2010?

(A) $700 \left(\frac{100 + n}{100} \right)$

(B) $700 \left(\frac{100}{100 + n} \right)$

(C) $700 \left(\frac{100 - n}{100} \right)$

(D) $700 \left(\frac{100}{100 - n} \right)$

Answer

Let x be the number of houses in Town B.
Town A had $n\%$ more than Town B, so:

$$700 = x \left(\frac{100 + n}{100} \right) \Rightarrow x = 700 \cdot \left(\frac{100}{100 + n} \right)$$

Correct answer: (B) $700 \left(\frac{100}{100 + n} \right)$

Percent Relationship Problem

The sum of positive numbers a and b is 4.76 times the positive number c , and a is 124% of b . What percent of c is a ?

- (A) 216.85%
- (B) 238.00%
- (C) 263.50%
- (D) 295.12%

Answer

We are given:

$$a + b = 4.76c \quad \text{and} \quad a = 1.24b$$

Substitute a into the sum:

$$1.24b + b = 4.76c \Rightarrow 2.24b = 4.76c \Rightarrow b = \frac{4.76c}{2.24}$$

Now substitute b into $a = 1.24b$:

$$a = 1.24 \cdot \frac{4.76c}{2.24} = \frac{1.24 \cdot 4.76}{2.24}c = 2.635c$$

Convert to percent:

$$a = 263.5\% \text{ of } c$$

Correct answer: (C) 263.50%

Percent Relationship Problem

The sum of positive numbers x and y is 3.5 times the positive number z , and x is 160% of y . What percent of z is x ?

- (A) 212.00%
- (B) 233.33%
- (C) 256.00%
- (D) 280.00%

Answer

We are given:

$$x + y = 3.5z \quad \text{and} \quad x = 1.6y$$

Substitute x into the sum:

$$1.6y + y = 3.5z \Rightarrow 2.6y = 3.5z \Rightarrow y = \frac{3.5z}{2.6}$$

Now substitute y into $x = 1.6y$:

$$x = 1.6 \cdot \frac{3.5z}{2.6} = \frac{1.6 \cdot 3.5}{2.6}z = \frac{5.6}{2.6}z \approx 2.1538z$$

Convert to percent:

$$x \approx 215.38\% \text{ of } z$$

Closest answer: (A)212.00%

Thank You!

We appreciate your time and focus.

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